

Integration — Lesson 16 Test: Integrating Sine & Cosine

10 questions · show your work in the box · exact values (use π , not decimals) · don't forget + C on indefinite integrals

Name: _____ Date: _____ Score: _____ / 10

1. From the value table: what are $\sin(3\pi/2)$ and $\cos(2\pi)$?

1 pt

2. Find $\int \cos x \, dx$, and check your answer by differentiating it.

1 pt

3. Find $\int \sin x \, dx$. Explain in one sentence why the minus sign must appear in the answer.

1 pt

4. Find $\int 5 \sin x \, dx$.

1 pt

5. Find $\int (\cos x - \sin x) dx$, and check by differentiating.

1 pt

6. Compute $\int_0^{\pi/2} \sin x dx$. Show the evaluation at both limits, tracking every sign.

1 pt

7. Compute $\int_0^{\pi} \cos x dx$, and explain the answer using signed area (where is $\cos x$ above the axis, and where below?).

1 pt

8. Compute $\int_{\pi}^{2\pi} \sin x dx$. What does the sign of your answer tell you about the sine curve on this interval?

1 pt

9. Find the average value of $\cos x$ on the interval $[0, \pi/2]$. Give the exact answer.

1 pt

10. Challenge: compute $\int_0^\pi (2 \sin x + x) dx$ with full working.

1 pt
